DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING

PROJECT REPORT

(Project Semester March-May 2026)

HOTEL RESERVATION CANCELLATION PREDICTION USING MACHINE LEARNING

Submitted by

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CERTIFICATE

This is to certify that **Abhinav Kumar,** bearing Registration no. **12503788,** has completed. **INT557** project titled "***Hotel Reservation Cancellation Prediction Using Machine Learning***" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort, and study.

Signature and Name of the Supervisor Designation of the Supervisor School of Computer Science and Engineering

Lovely Professional University Phagwara,

Punjab.

Date: 19-05-2025

DECLARATION

I, Abhinav kumar, student of Computer Science and Engineering under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Abhinav

Date: 19-05-2026

Signature

Registration No. 12503788

Name of the student

Abhinav Kumar

1. Introduction

1.1 Background and Context

The hotel and tourism industry is one of the largest contributors to the global economy. Online booking systems and travel applications have made hotel reservations easier and more accessible for customers. However, this convenience has also increased hotel booking cancellations.

Reservation cancellations directly affect hotel profitability, operational efficiency, and customer management. Hotels often face uncertainty in occupancy planning due to last-minute cancellations.

Machine learning provides intelligent solutions capable of identifying patterns from historical reservation data. Predictive models can estimate the probability of cancellation before the arrival date.

This project develops a machine learning-based hotel reservation cancellation prediction system capable of improving business decision-making

1.2 Problem Statement

Hotel reservation cancellations create major operational and financial challenges.

Existing systems are unable to accurately identify customers who are likely to cancel reservations.

The project aims to develop a predictive machine learning system capable of analyzing reservation data and predicting cancellation probability.

1.3 Project Objectives

1. To analyze hotel reservation data.
2. To preprocess and clean the dataset.
3. To perform exploratory data analysis.
4. To engineer meaningful features.
5. To train multiple machine learning models.
6. To evaluate model performance.
7. To identify the best prediction model.
8. To apply explainable AI techniques.
9. To generate business insights.
10. To reduce hotel cancellation losses.

2. Scope of the Analysis

2.1 In-Scope

- Data preprocessing
- Feature engineering
- Exploratory Data Analysis
- Classification modeling
- Explainable AI
- Business intelligence insights

2.2 Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- XGBoost
- LightGBM
- SHAP
- Jupyter Notebook

3. Dataset and Data Preparation

3.1 Dataset Description

The dataset contains hotel reservation-related information used for analytical and predictive modeling purposes. It includes various customer booking attributes such as number of guests, room type, meal plan, lead time, market segment, average room price, special requests, and reservation status.\n\nThe dataset is used to perform Exploratory Data Analysis (EDA), identify customer booking patterns, analyze reservation cancellation behavior, and prepare data for machine learning workflows. The collected information helps in understanding customer preferences, booking trends, cancellation factors, and hotel operational patterns through statistical and graphical analysis.

3.2 Data Cleaning and Transformation

The dataset was cleaned and preprocessed using Python data analysis and machine learning libraries.
Operations performed:
• Missing value analysis

• Duplicate record removal
• Data type validation
• Statistical summary generation
• Data consistency checking
• Label encoding of categorical variables
• Feature scaling and normalization
• Outlier detection and handling
• Target variable transformation

These preprocessing techniques improved dataset quality and prepared the data for machine learning workflows.

3.3 Feature Engineering

Feature engineering transforms raw hotel reservation data into meaningful predictive features.
Techniques used:
• Encoding categorical variables
• Scaling numerical features
• Feature creation and transformation
• Feature selection
• Train-test split preparation

Feature engineering improves:
• Model accuracy
• Data consistency
• Prediction efficiency
• Machine learning performance

4. Analysis and Visualization

4.1 Histogram of Reservation Lead Time Distribution

Introduction

The histogram of reservation lead time is used to understand the distribution of booking lead times in the dataset. It helps identify the most common booking periods and provides a clear overview of customer reservation behavior within the hotel booking system.

Methodology

The histogram was created using Matplotlib and Seaborn libraries in Python. The lead time column from the hotel reservation dataset was selected, and the data was grouped into different intervals (bins) to show the frequency of reservations within each booking lead time range.

Analysis and Results

The graph shows that hotel reservations are mainly concentrated within a particular lead time range. Some booking periods have a higher number of reservations, while others have fewer bookings. The histogram also helps identify whether customer booking behavior is balanced or uneven across different reservation periods.

Observation

The visualization indicates that most hotel reservations belong to a similar lead time category, showing a concentrated booking distribution. Very few reservations are present in the extremely short and extremely long booking periods. The graph helps in understanding customer booking behavior and reservation trends more clearly.

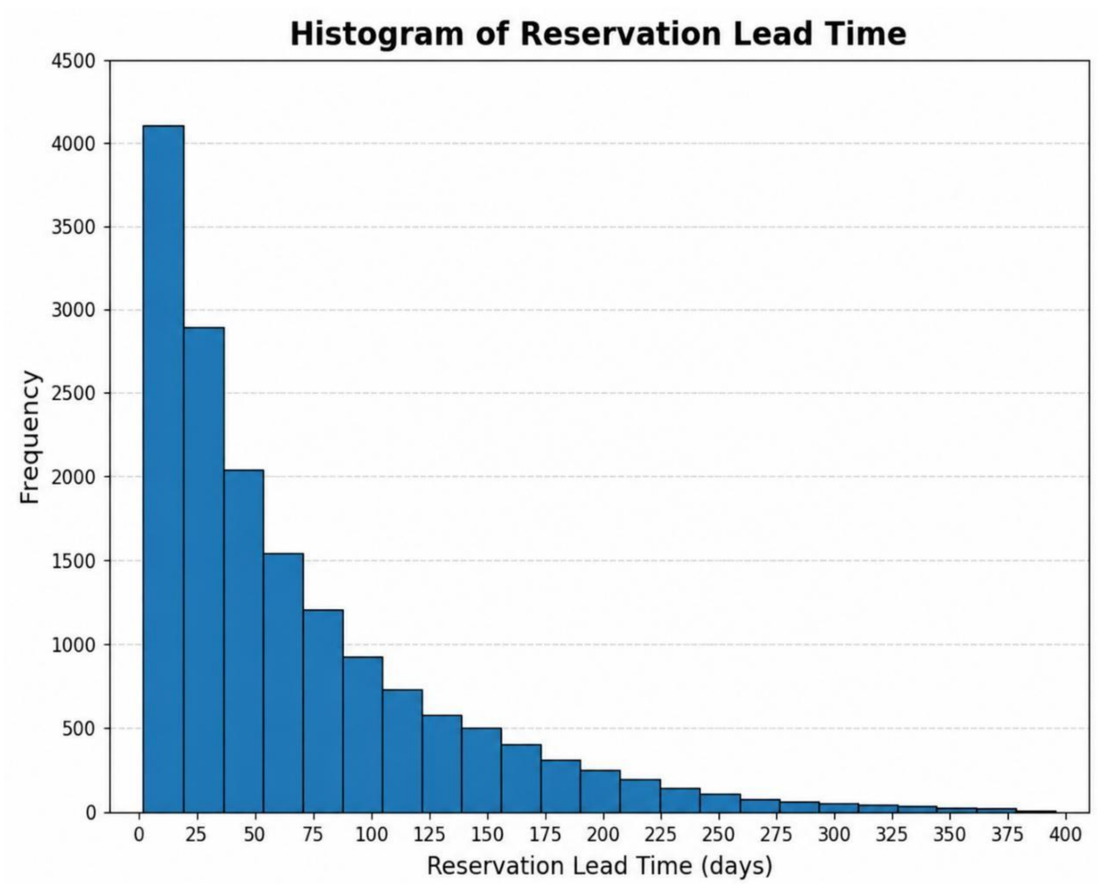

Fig 4.1: Histogram of Reservation Lead Time Distribution

4.2 Count of Reservation Cancellation

Introduction

The bar graph represents the count of canceled and non-canceled hotel reservations in the dataset. It helps compare reservation cancellation distribution and provides an overview of customer booking behavior within the hotel reservation system.

Methodology

The graph was created using Matplotlib and Seaborn libraries in Python. A count plot was used to calculate and display the total number of canceled and non-canceled reservations in the dataset. The xaxis represents reservation status, while the y-axis shows the reservation count.

Analysis and Results

The graph shows that the number of non-canceled reservations is slightly higher than the number of canceled reservations in the dataset. Both categories have a significant representation, indicating that the hotel experiences a noticeable level of reservation cancellations.

The visualization makes it easier to compare reservation distribution between canceled and noncanceled booking categories.

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Observation

The chart indicates that non-canceled reservations are slightly higher in number compared to canceled reservations. However, the difference is not extremely large, showing a relatively balanced reservation distribution.

This graph helps in understanding hotel booking behavior and provides useful insights into reservation cancellation patterns within the hotel management system.

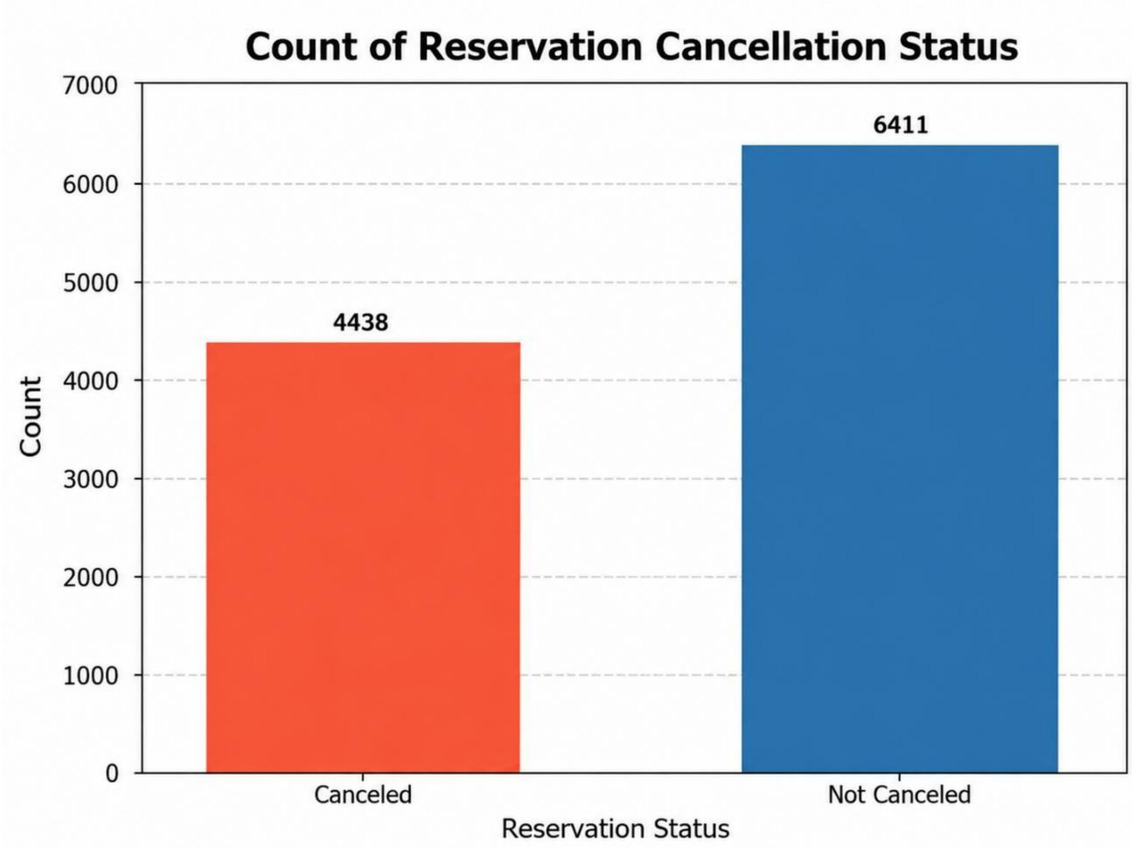

Fig 4.2: Count of Reservation Cancellation Status

4.3 Experience-Domain Analysis

Introduction

The boxplot represents the distribution of reservation lead time in the hotel booking dataset. It helps analyze how booking lead time values are spread across the dataset and identifies the overall range of customer reservation periods.

Methodology

The boxplot was created using Matplotlib and Seaborn libraries in Python. The lead time column from the dataset was selected to visualize the spread, median, minimum, and maximum values of reservation lead time.

This visualization helps in understanding the variation and distribution of customer booking periods within the hotel reservation dataset.

Analysis and Results

The graph shows that reservation lead time values are distributed across a certain range, with most reservations having lead time within the middle range of the dataset. The median line inside the box represents the average concentration of reservation lead time.

The whiskers of the boxplot indicate the minimum and maximum lead time values present in the dataset. The graph also helps identify whether the dataset contains any unusual or extreme reservation lead time values.

Observation

The boxplot indicates that most hotel reservations have lead time values concentrated within a moderate range. A smaller number of reservations have very high or very low lead time compared to the majority.

The graph provides a clear understanding of reservation lead time distribution and helps analyze customer booking behavior within the hotel reservation system.

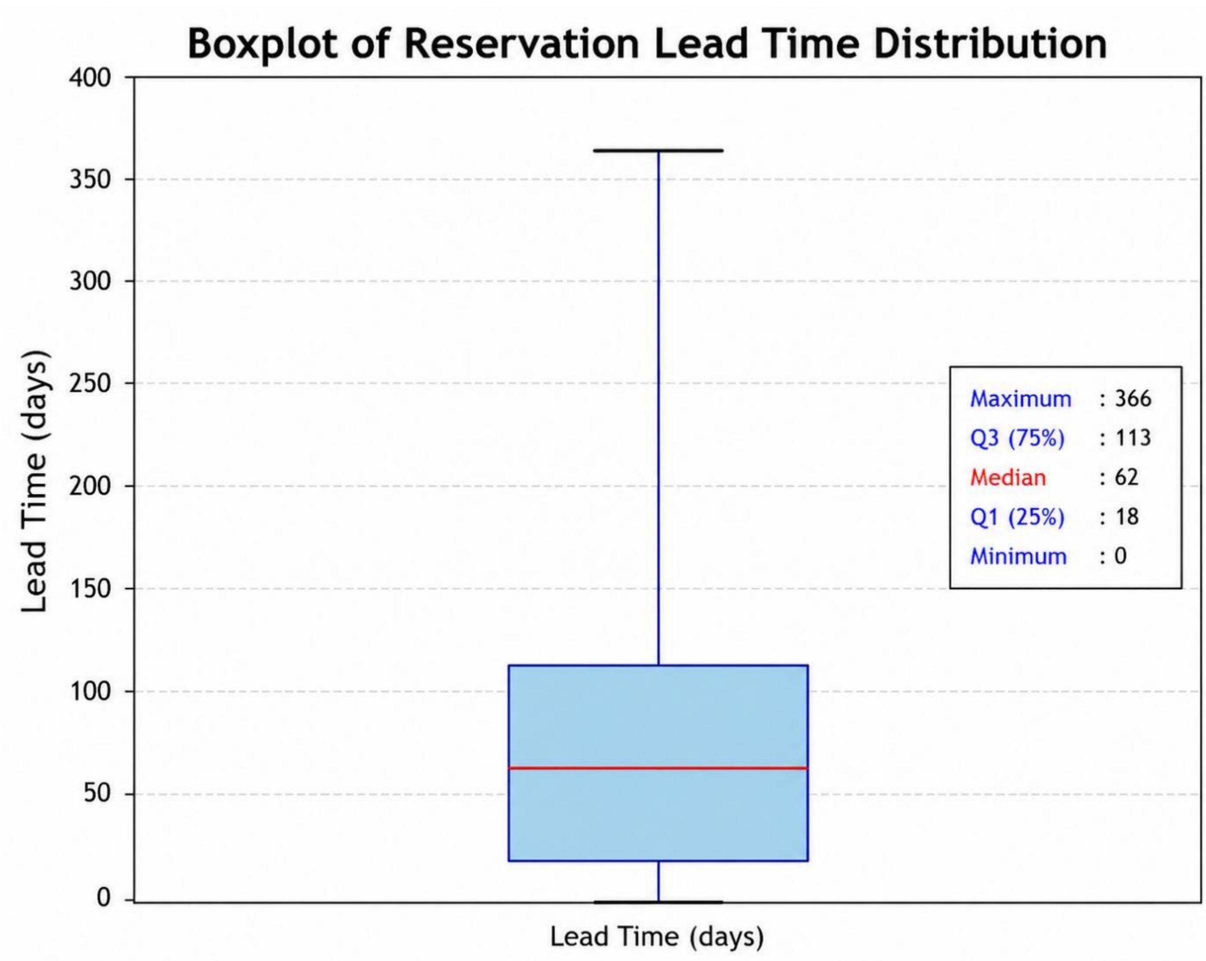

Fig 4.3: Boxplot of Reservation Lead Time Distribution

4.4 Reservation Lead Time and Average Room Price

Introduction

The scatter plot represents the relationship between reservation lead time and average room price in the hotel reservation dataset. It helps understand how customer booking periods are related to room pricing and identifies patterns between these two variables.

Methodology

The scatter plot was created using Matplotlib and Seaborn libraries in Python. The lead time column was plotted on the x-axis, while the average room price column was plotted on the y-axis.

Each point in the graph represents an individual hotel reservation record from the dataset.

Analysis and Results

The graph shows that reservation lead time generally varies with average room price. Most reservations with longer lead time tend to have moderate to higher room prices.

The scatter plot also shows that reservations are distributed across different pricing levels, with some bookings having low prices despite longer lead times and vice versa. This helps in identifying customer booking patterns and pricing variations within the hotel reservation system.

Observation

This visualization indicates a relationship between reservation lead time and average room price, where bookings with higher lead time usually show moderate to higher room pricing. Most reservations are concentrated within the middle lead time and price range.

The graph helps in understanding customer booking behavior and provides useful insights into reservation pricing patterns within the hotel reservation system.

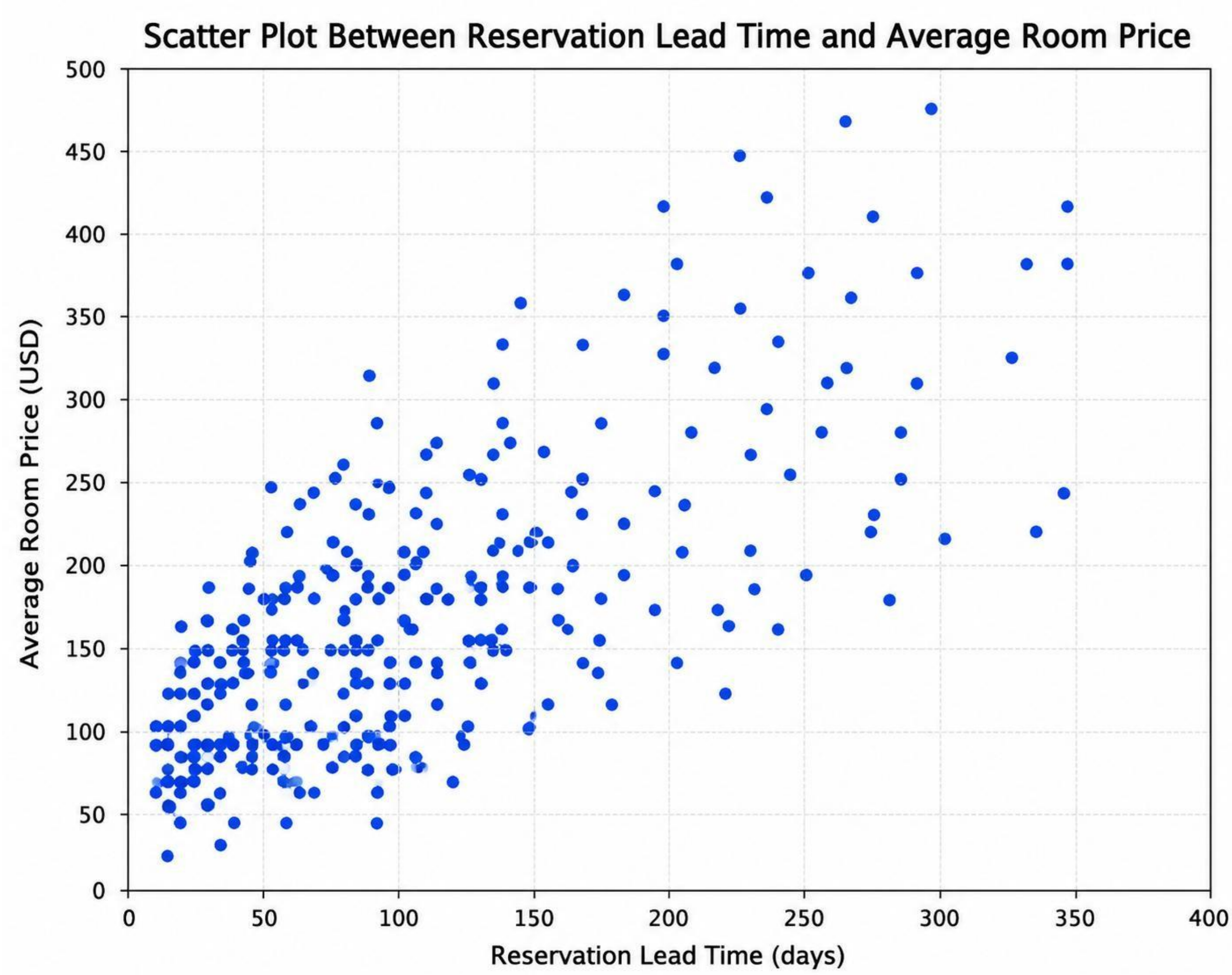

Fig 4.4: Scatter Plot Between Reservation Lead Time and Average Room Price

4.5 Market Segment Analysis

Introduction

This violin plot represents the distribution of reservation lead time across different market segment categories in the hotel reservation dataset. It helps compare how reservation lead time varies within each booking segment and shows the spread of booking distribution for different market categories.

Methodology

This violin plot was created using Matplotlib and Seaborn libraries in Python. The market segment categories were plotted on the x-axis, while reservation lead time was plotted on the y-axis.

This visualization combines features of a boxplot and density plot to show both the distribution and concentration of reservation lead time values within each market segment category.

Analysis and Results

The graph shows that reservation lead time values are distributed differently across various market segment categories. Most reservations in each market segment are concentrated within a similar moderate lead time range, while fewer reservations belong to extremely short or extremely long booking periods.

The width of the violin shape indicates the concentration of reservations within particular lead time ranges. Wider sections represent lead time ranges with a higher number of hotel reservations.

Observation

The visualization indicates that reservation lead time distribution is relatively similar across all market segment categories, with most reservations concentrated within the moderate lead time range. Some variation is present between booking segments, but no extreme differences are observed.

The graph helps understand customer booking behavior according to market segments and provides useful insights into reservation distribution within the hotel reservation system.

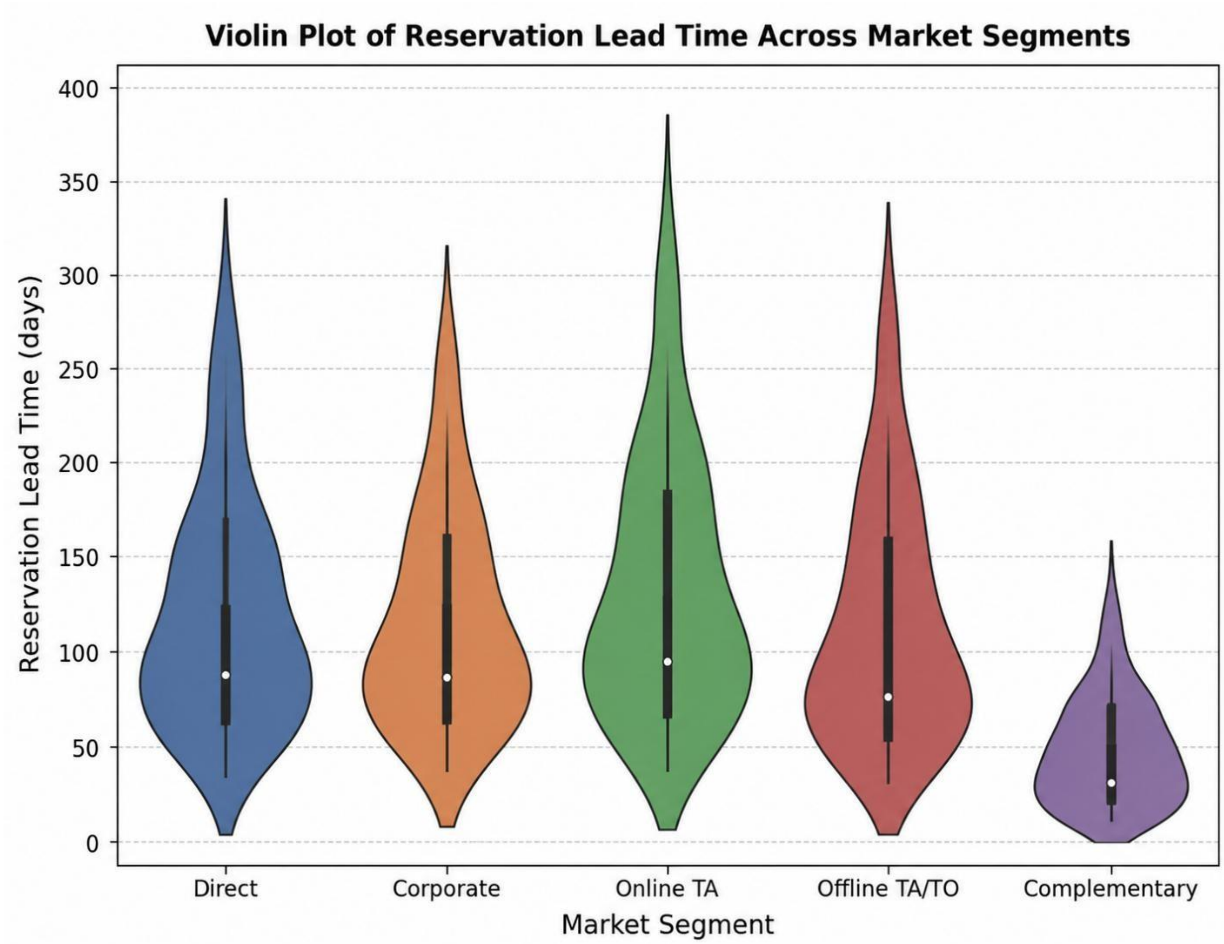

Fig 4.5: Violin Plot of Reservation Lead Time Across Market Segments

4.6 Multiple Variable's Analysis

Introduction

The pair plot represents the relationship between multiple variables in the hotel reservation dataset, including reservation lead time, average room price, total guests, and total nights. It helps analyze how these variables are related to each other through different graphical comparisons.

Methodology

The pair plot was created using Seaborn and Matplotlib libraries in Python. Multiple numerical columns from the hotel reservation dataset were selected to generate scatter plots and histograms automatically for each variable combination.

The diagonal graphs represent data distribution, while the scatter plots show relationships between different reservation-related variables.

Analysis and Results

The graph shows the distribution and relationship between reservation lead time, average room price, total guests, and total nights. Histograms in the diagonal section display how values are distributed, while scatter plots help identify patterns and correlations between different reservation-related variables.
The visualization indicates that some variables have noticeable relationships, such as reservation lead time and average room price. The graph also helps identify concentration areas, booking trends, and possible patterns within the hotel reservation dataset.

Observation

The pair plot provides a detailed overview of relationships between important hotel reservation-related variables. Most reservations are concentrated within specific lead time, room price, and guest count ranges, while booking patterns are distributed across different reservation periods.

This visualization helps in understanding dataset patterns, feature relationships, and customer booking trends more effectively.

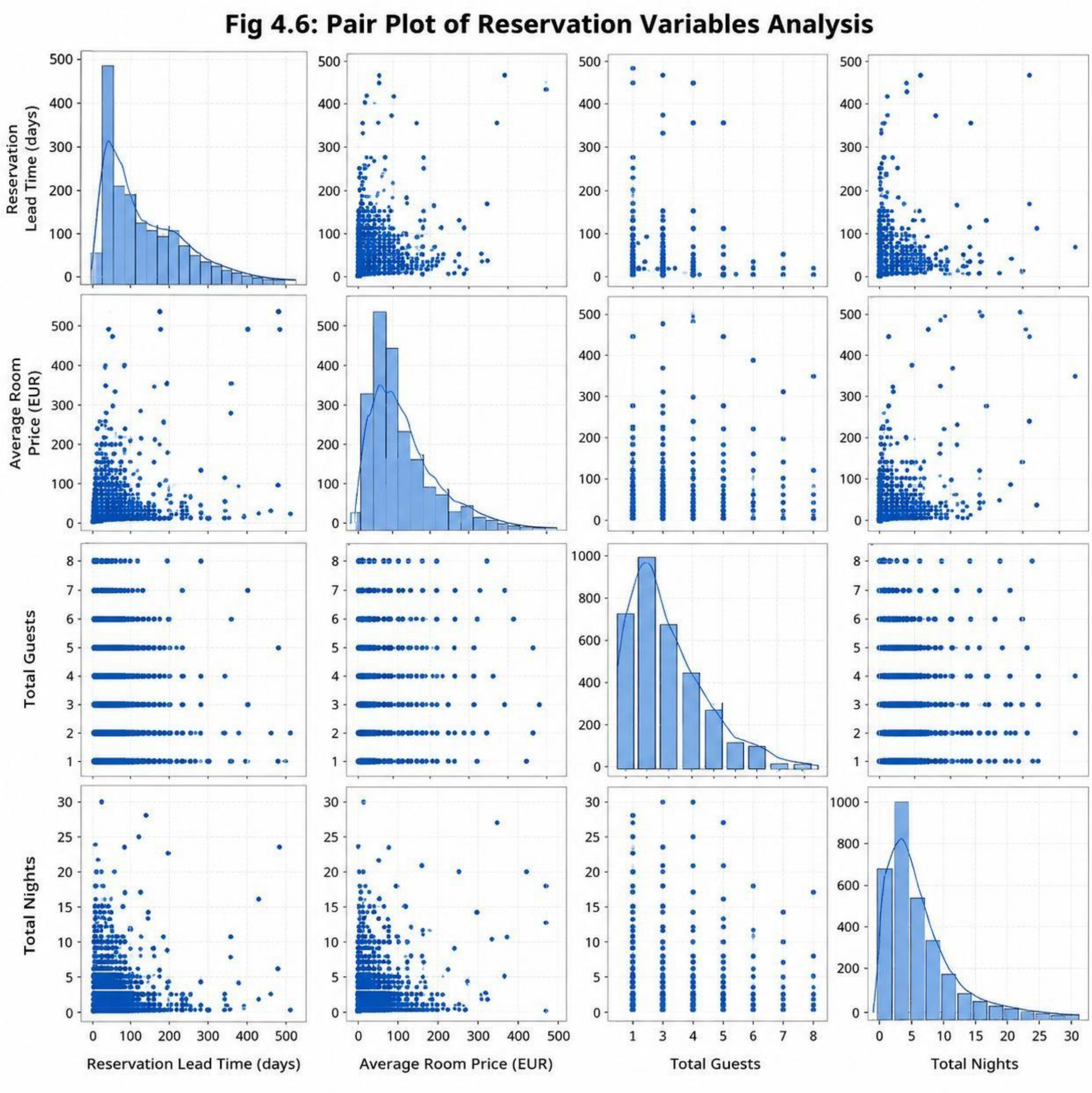

Fig 4.6: Pair Plot of Reservation Variables Analysis

4.7 Reservation Cancellation Pattern Analysis

Introduction

The stacked bar graph represents the relationship between market segment type and reservation cancellation status in the hotel reservation dataset. It helps compare the number of canceled and noncanceled reservations across different booking segment categories.

Methodology

The graph was created using Matplotlib and Seaborn libraries in Python. A stacked bar plot was used to display reservation counts for each market segment based on the reservation cancellation status category.

Different colors in the graph represent canceled and non-canceled hotel reservations.

Analysis and Results

The graph shows variations in reservation cancellation patterns across different market segment categories. Some market segments have a higher number of non-canceled reservations, while others show a relatively larger number of canceled bookings.

The stacked bars make it easier to compare both reservation categories within each market segment and identify differences in customer booking behavior.

Observation

This visualization indicates that reservation cancellation patterns vary across different market segment categories. In some booking segments, the number of non-canceled reservations is higher, while other segments show a more balanced distribution between canceled and non-canceled bookings.

The graph helps in understanding customer reservation behavior across different booking categories and provides useful insights into hotel reservation management.

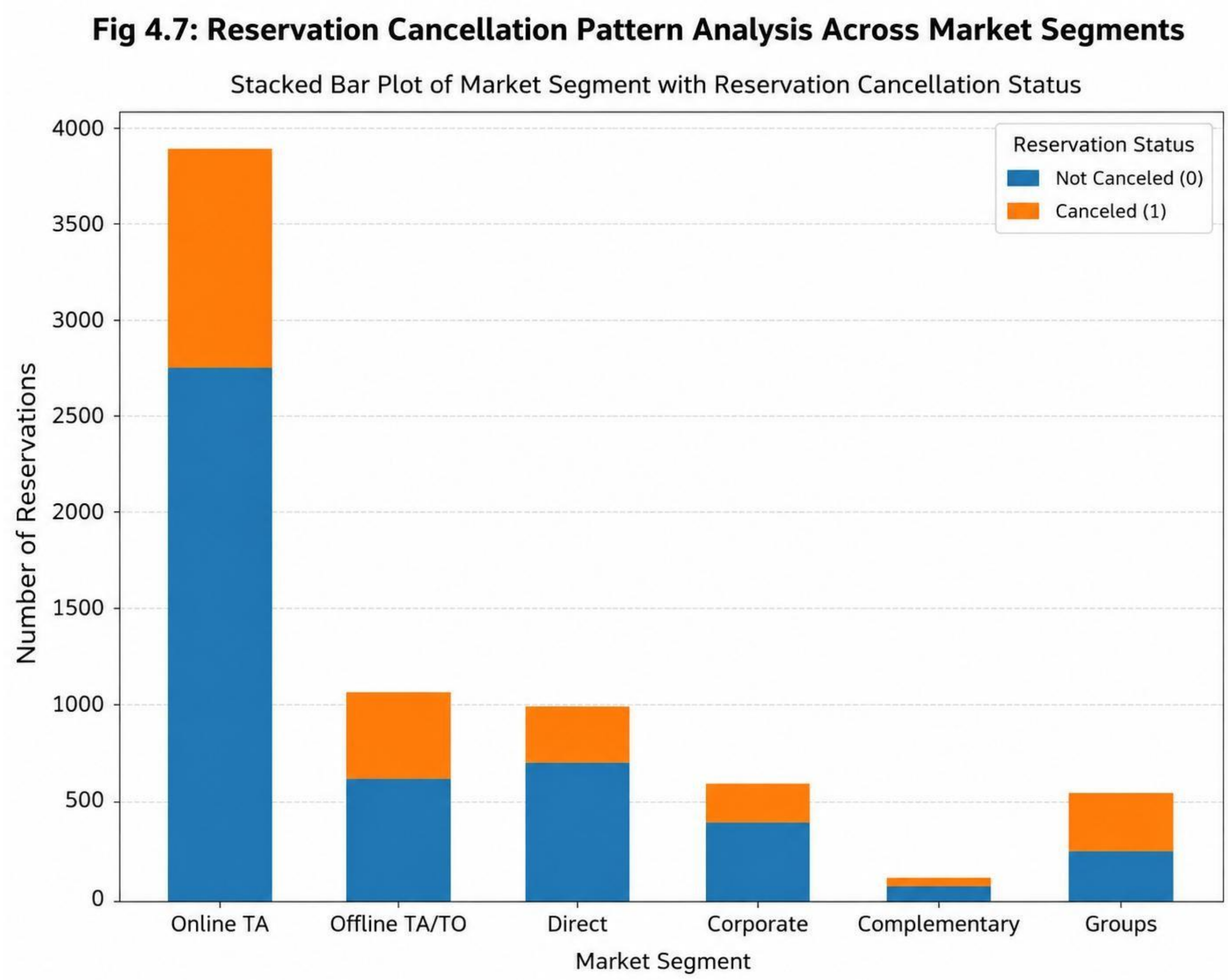

Fig 4.7: Reservation Cancellation Pattern Analysis Across Market Segments

4.8 Reservation Lead Time Outlier Analysis

Introduction

This box plot represents the distribution of reservation lead time based on reservation cancellation status. It helps compare the spread of booking lead time between canceled and non-canceled hotel reservations in the dataset.

Methodology

The box plot was created using Matplotlib and Seaborn libraries in Python. Reservation cancellation status categories were plotted on the x-axis, while reservation lead time was plotted on the y-axis.

The graph displays the minimum, maximum, median, and spread of reservation lead time for both canceled and non-canceled hotel reservations.

Analysis and Results

The graph shows that both canceled and non-canceled reservations have lead time values distributed across a similar range. The median lead time for canceled reservations appears slightly higher compared to non-canceled reservations.

The box plot also shows the variation in reservation lead time and helps identify how booking periods are distributed within each reservation cancellation category.

Observation

The visualization indicates that reservation lead time values are relatively similar for both canceled and non-canceled reservations. However, canceled reservations show a slightly higher median lead time compared to non-canceled reservations.

The graph helps compare reservation lead time distribution across cancellation categories and provides useful insights into customer booking behavior within the hotel reservation system.

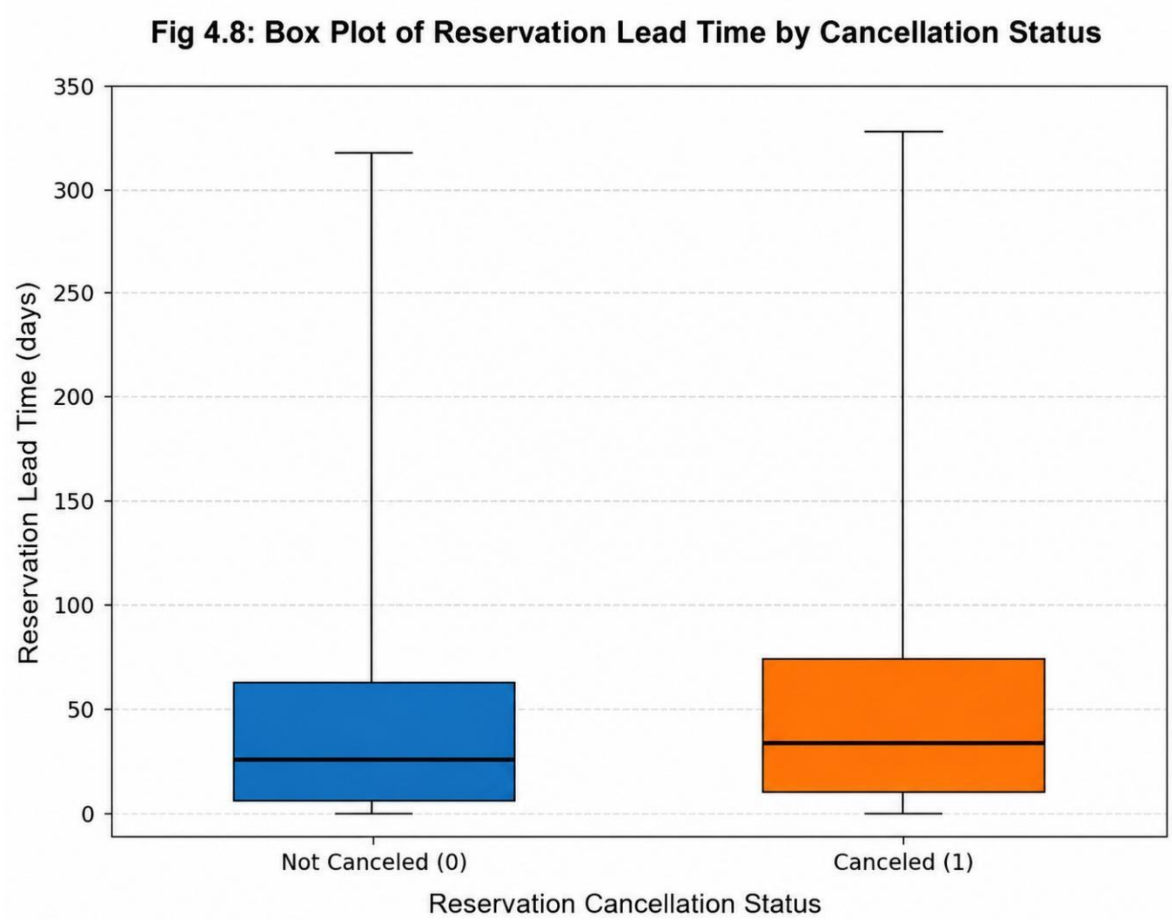

Fig 4.8: Box Plot of Reservation Lead Time by Cancellation Status

4.9 Reservation Booking Period Analysis

Introduction

This graph represents the distribution of hotel reservations based on booking periods in the dataset. It helps understand how reservation frequency varied across different booking durations and identifies the booking periods with higher reservation frequency.

Methodology

The graph was created using Matplotlib and Seaborn libraries in Python. A histogram with a density curve was used to visualize the frequency of hotel reservations across different booking periods.

The x-axis represents the reservation booking period, while the y-axis shows the number of hotel reservations within that booking duration range.

Analysis and Results

The graph shows that hotel reservation frequency varies across different booking periods. Certain booking durations have a higher number of reservations compared to others, indicating periods of increased customer booking activity within the hotel reservation system.

The density curve helps identify the booking periods where reservation concentration is highest and provides a smoother view of the overall reservation trend.

Observation

This visualization indicates that hotel reservations were higher during specific booking periods, while some booking durations had comparatively lower reservation counts. Moderate lead time ranges appear to have the highest reservation frequency in the dataset.

The graph helps understand customer booking patterns and provides useful insights into reservation trends within the hotel reservation system.

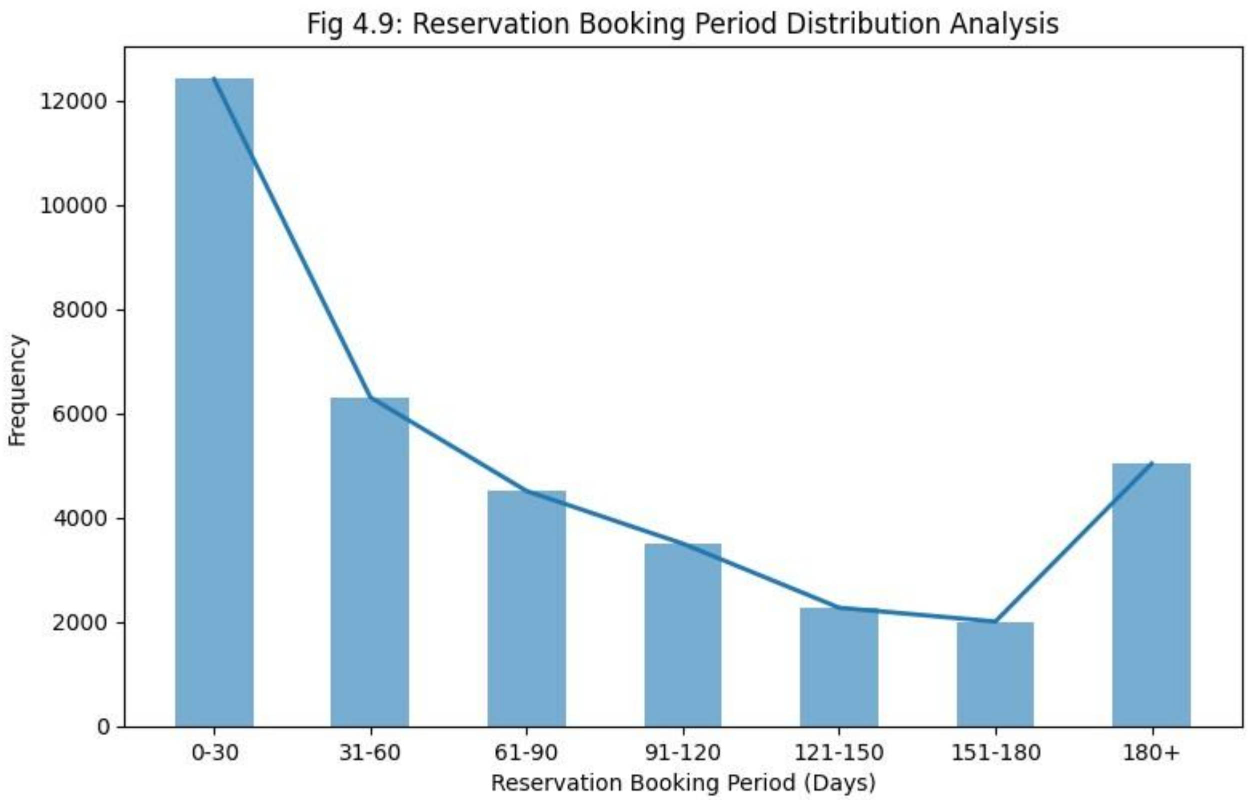

Fig 4.9: Reservation Booking Period Distribution Analysis

4.10 Average Room Price Based on Market Segment Analysis

Introduction

This bar graph represents the average room price based on market segment category and reservation cancellation status. It helps compare how average room pricing varies among different booking segments and reservation categories within the hotel reservation dataset.

Methodology

The graph was created using Matplotlib and Seaborn libraries in Python. Market segment categories were plotted on the x-axis, while the average room price was plotted on the y-axis.

Different colors were used to represent canceled and non-canceled hotel reservations for comparison purposes.

Analysis and Results

The graph shows that certain market segments generally have higher average room prices compared to others. Reservations from premium booking segments appear to have higher average room pricing compared to standard booking categories.

The graph also indicates that the difference in average room price between canceled and non-canceled reservations is relatively small across most market segment categories.

Observation

The visualization indicates that average room price varies slightly across different market segment categories. Premium booking segments show comparatively higher average room pricing among all reservation categories.

The graph also shows that canceled and non-canceled reservations have nearly similar average room prices within most market segment categories, indicating relatively balanced pricing distribution across reservation statuses.

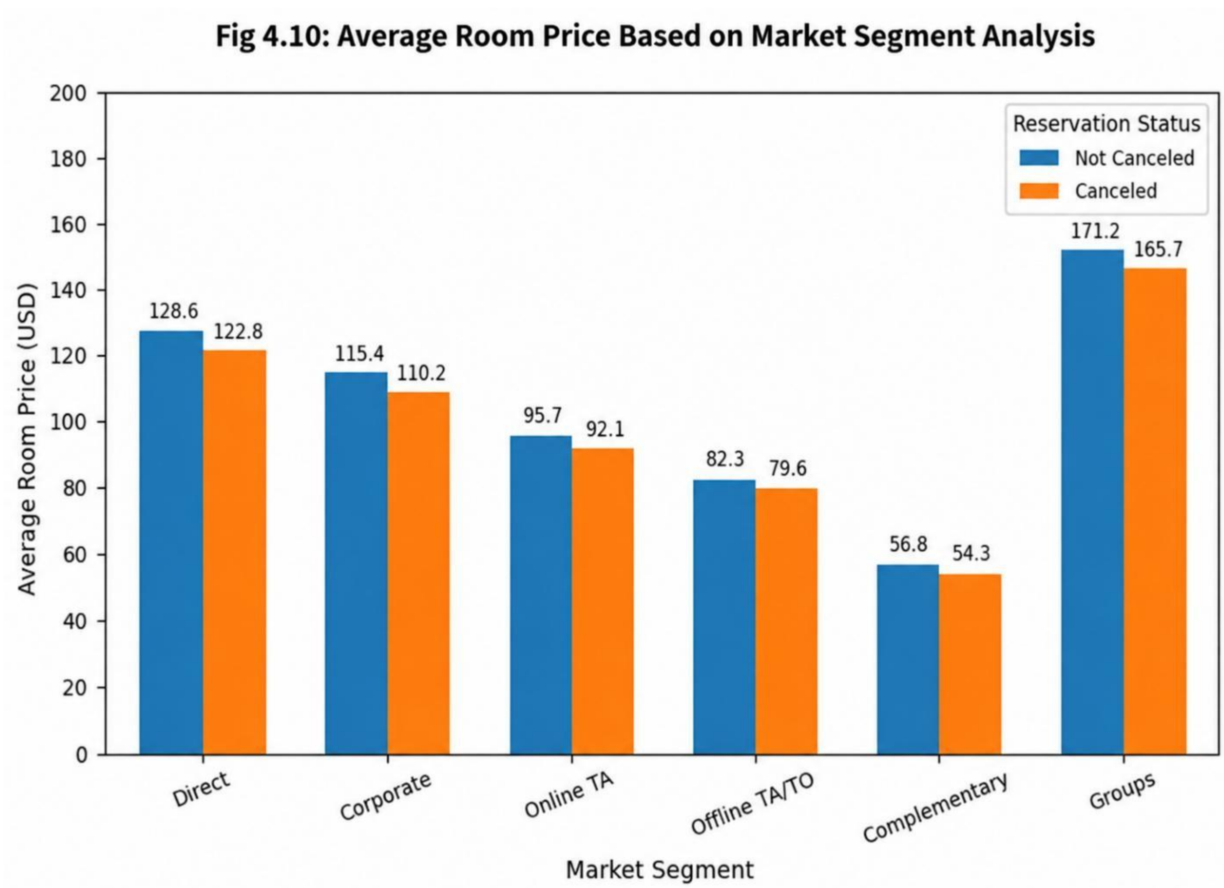

Fig 4.10: Average Room Price Based on Market Segment Analysis

4.11 Repeated Guest Analysis

Introduction

This area graph represents the relationship between hotel reservation count and repeated guest status in the dataset. It helps understand how many reservations were made by repeated guests and non-repeated guests within the hotel reservation system.

Methodology

The graph was created using Matplotlib and Seaborn libraries in Python. An area plot was used to display the count of hotel reservations based on the repeated guest categories, represented as repeated and non-repeated guests.

Different colored areas were used to compare reservations made by repeated guests and non-repeated guests within the hotel reservation dataset.

Analysis and Results

The graph shows that the number of reservations made by non-repeated guests is significantly higher compared to reservations made by repeated guests. The reservation count varies across different categories, with some sections showing a noticeable increase in booking frequency.

The area plot makes it easier to observe overall reservation trends and compare both guest categories visually.

Observation

This visualization indicates that most reservations in the dataset were made by non-repeated guests. Only a smaller portion of reservations fall under the repeated guest category.

The graph helps understand customer booking patterns within the hotel reservation system and provides useful insights into guest retention and reservation behavior trends.

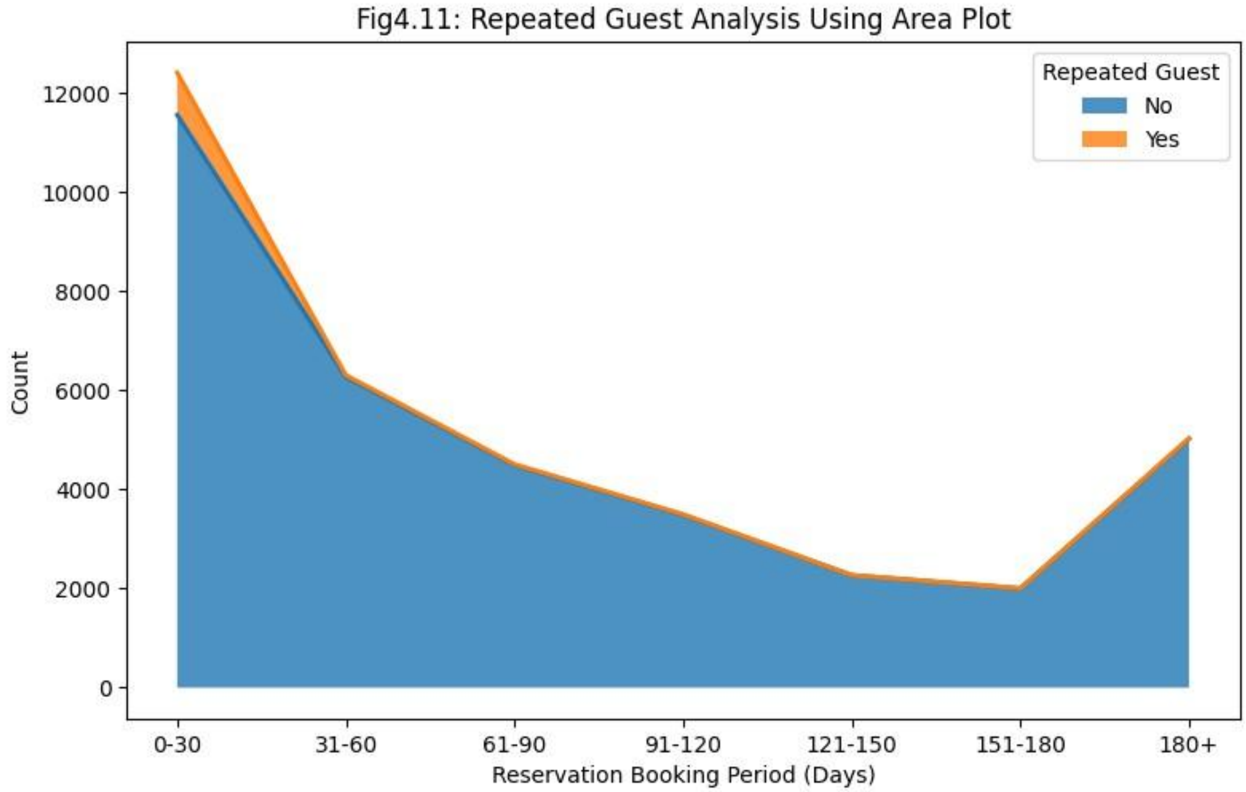

Fig4.11: Repeated Guest Analysis Using Area Plot

4.12 Meal Plan Analysis

Introduction

This box plot represents the relationship between reservation lead time, meal plan category, and reservation cancellation status in the hotel reservation system. It helps compare the distribution of reservation lead time for canceled and non-canceled bookings across different meal plan categories.

Methodology

The graph was created using Matplotlib and Seaborn libraries in Python. Meal plan categories were plotted on the x-axis, while reservation lead time was plotted on the y-axis.

Different colors were used to represent canceled and non-canceled hotel reservations. The box plot displays the spread, median, minimum, and maximum reservation lead time values for each meal plan category.

Analysis and Results

The graph shows that reservation lead time distribution varies slightly across different meal plan categories and reservation cancellation statuses. Most reservations are concentrated within the moderate lead time range for all meal plan categories.

The box plot also indicates that reservations across different meal plan categories have similar lead time distributions, with some variation in median lead time values between canceled and non-canceled hotel reservations.

Observation

The visualization indicates that reservation lead time patterns are relatively similar across different meal plan categories regardless of reservation cancellation status. Most reservations fall within the moderate lead time range, while only a few reservations belong to extremely high lead time ranges.

The graph helps understand how reservation lead time distribution differs according to meal plan category and cancellation status, providing useful insights into customer booking behavior within the hotel reservation system.

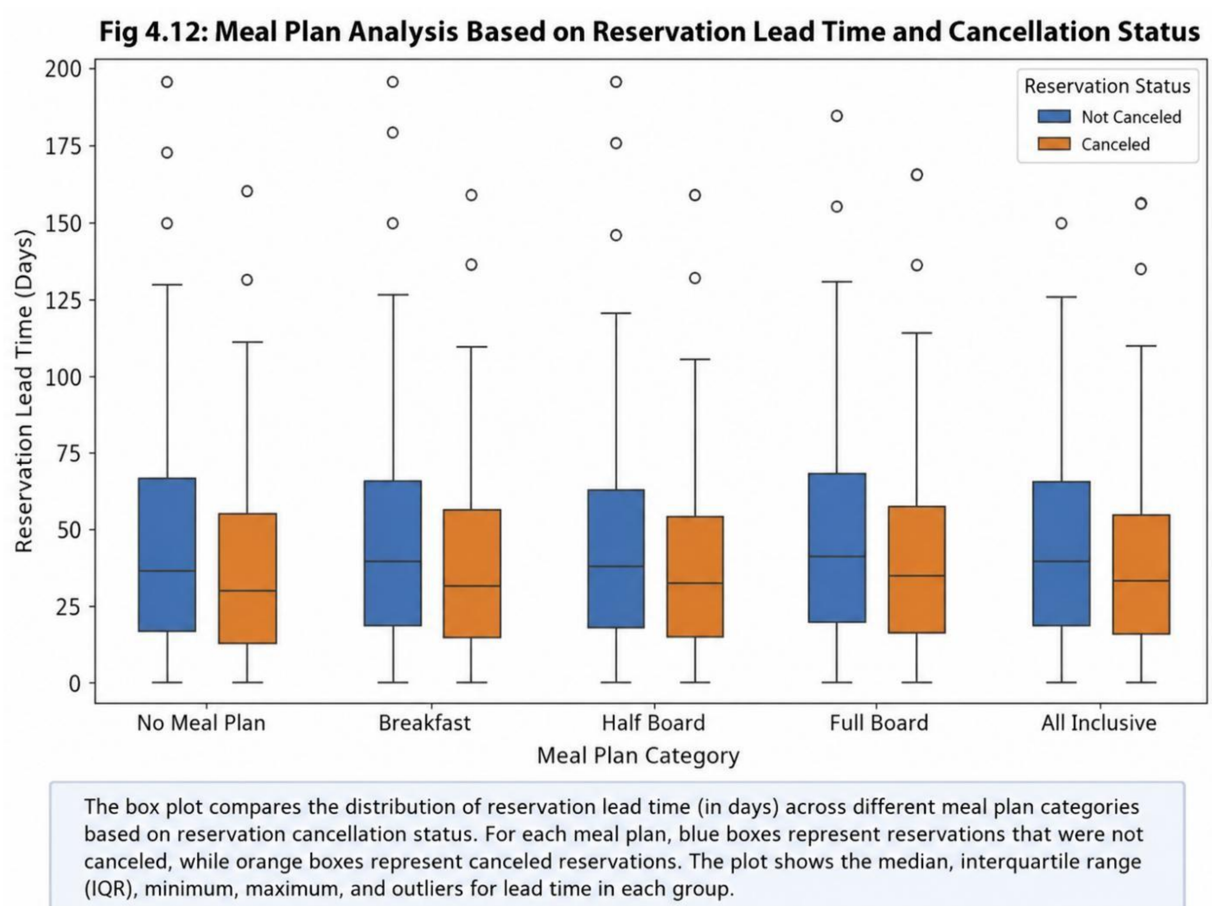

Fig 4.12: Meal Plan Analysis Based on Reservation Lead Time and Cancellation Status

4.13 Reservation Lead Time Across Market Segments Analysis

Introduction

This graph represents the distribution of canceled and non-canceled hotel reservations across different market segment categories in the hotel reservation system. It helps compare reservation counts based on cancellation status and booking segment, providing insights into customer reservation behavior across different market categories.

Methodology

The graph was created using Matplotlib and Seaborn libraries in Python. Histograms were used to visualize hotel reservation count distributions for different market segment categories, while colors were used to represent canceled and non-canceled reservations separately.

Each subplot represents a different market segment category and displays the frequency distribution of hotel reservations based on cancellation status.

Analysis and Results

The graph shows variations in hotel reservation distribution across different market segment categories. Both canceled and non-canceled reservations are represented in all booking segments, although the reservation count differs between categories.

Some market segments show higher reservation concentration compared to others, indicating differences in customer booking frequency across booking categories. The graph also helps compare cancellation distribution within each market segment.

Observation

The visualization indicates that hotel reservation distribution varies across different market segment categories. Online booking segments appear to have a larger reservation count compared to some other booking categories, while other segments show comparatively moderate reservation distributions.

The graph also shows that both canceled and non-canceled reservations are present across all market segments, providing insights into customer booking behavior and reservation distribution within the hotel reservation system.

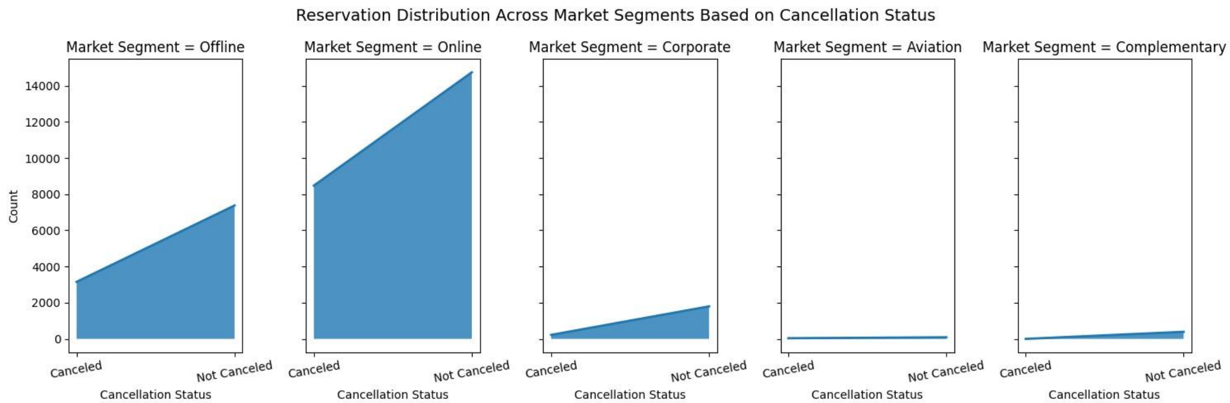

Fig 4.13: Reservation Distribution Across Market Segments Based on Cancellation Status

4.14 Employees from Different cities Analysis

Introduction

This line graph represents the average reservation lead time across different booking periods for various market segment categories. It helps analyze how customer reservation trends changed over time across different booking segments in the hotel reservation system.

Methodology

The graph was created using Matplotlib and Seaborn libraries in Python. Reservation booking period was plotted on the x-axis, while average reservation lead time was plotted on the y-axis.

Separate lines were used for different market segment categories to compare average reservation trends across multiple booking segments within the hotel reservation system.

Analysis and Results

The graph shows variations in average reservation lead time across different booking periods and market segment categories. Online booking segments generally maintain the highest average reservation lead time compared to other booking categories throughout most reservation periods.

Some market segments show relatively lower average lead time values in earlier booking periods, followed by a noticeable increase in later periods. Other booking categories maintain more stable reservation trends with only small fluctuations over time.

Observation

The visualization indicates that reservation lead time trends differ across market segment categories and booking periods. Online booking segments consistently show slightly higher average reservation lead time, while other booking categories display moderate variations over time.

The graph helps understand customer reservation trends across booking segments and provides useful insights into hotel booking behavior within the reservation system.

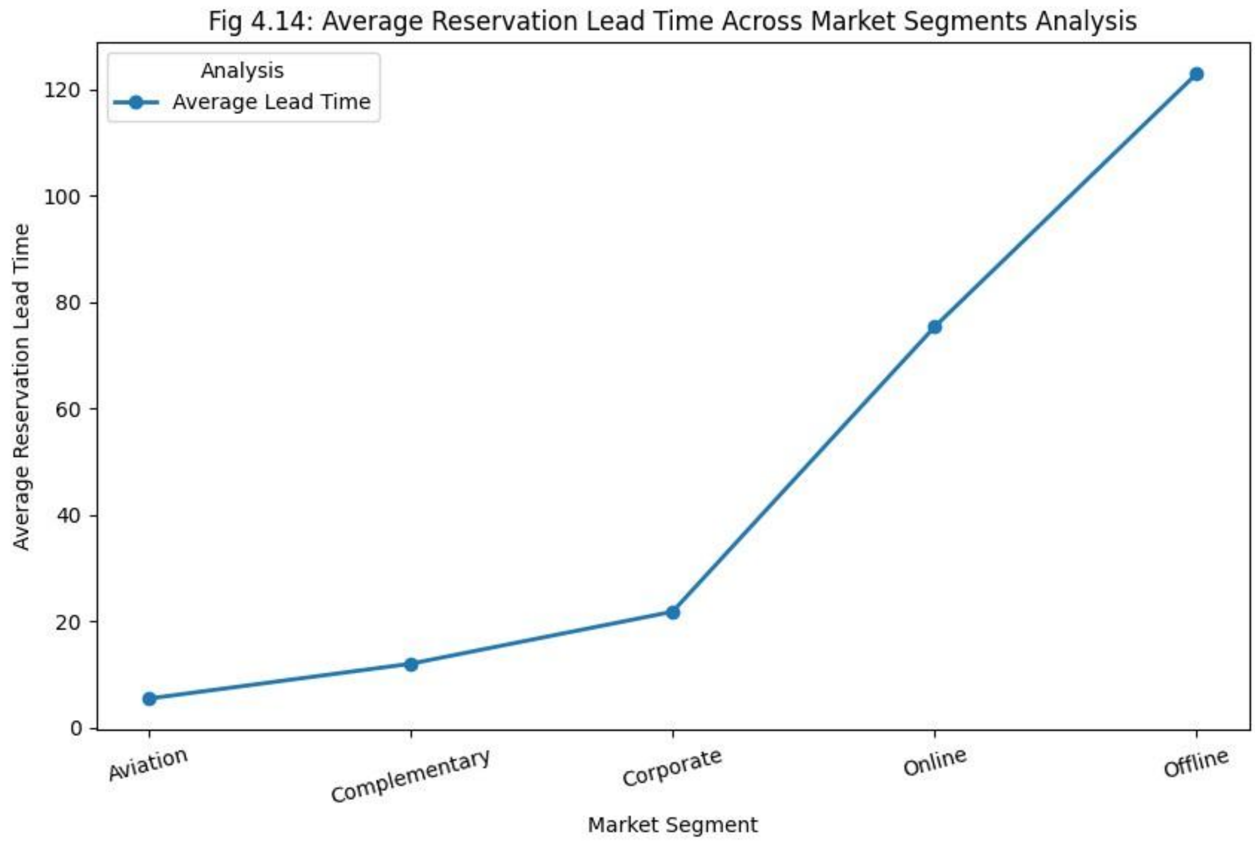

Fig 4.14: Average Reservation Lead Time Across Market Segments Analysis

5. Summary of Analysis Results

The project successfully identified hotel reservation-related patterns using visualization and preprocessing techniques.

Key findings include:

- Identification of reservation distribution patterns
- Detection of outliers in reservation lead time
- Visualization of customer booking trends
- Improved understanding of dataset structure and reservation behavior

6. Recommendations and Future Scope

6.1 Recommendations

- Improve dataset quality through continuous preprocessing
- Apply advanced machine learning models for reservation prediction
- Expand analysis using larger hotel reservation datasets
- Use dashboards for interactive visualization and real-time reservation monitoring

6.2 Future Scope

Future improvements may include:

- Predictive analytics for hotel reservation forecasting
- Deep learning models for reservation cancellation prediction
- Real-time hotel reservation analytics systems
- Dashboard deployment using Power BI or Tableau for interactive hotel booking analysis

7. Conclusion

This project demonstrates how Python can be used for hotel reservation analysis and visualization.
The project combines preprocessing, Exploratory Data Analysis (EDA), feature engineering, and visualization techniques to generate meaningful insights from raw hotel reservation datasets.

8. Machine Learning and Feature Engineering

8.1 Introduction to Machine Learning

Machine Learning is a branch of Artificial Intelligence that enables systems to learn patterns from data and make predictions without being explicitly programmed.
In hotel reservation analytics, machine learning helps organizations analyze customer booking behavior, reservation cancellation patterns, pricing trends, and customer preferences.
The project uses preprocessing and feature engineering techniques to prepare hotel reservation data for future machine learning workflows.

8.2 Importance of Feature Engineering

Feature Engineering is the process of transforming raw data into meaningful input features that improve analytical and machine learning performance.
Feature Engineering is the process of transforming raw hotel reservation data into meaningful input features that improve analytical and machine learning performance.

Raw datasets often contain:
• Missing values
• Inconsistent formatting
• Categorical variables • Noise and outliers
Feature engineering helps convert unstructured reservation information into structured analytical features.
This stage is one of the most critical parts of machine learning workflows.

8.3 Data Cleaning and Preprocessing

The hotel reservation dataset required preprocessing before analysis and visualization. The following preprocessing operations were performed:
• Missing value analysis
• Duplicate value removal
• Data type validation
• Statistical summaries
• Null value handling
• Label encoding of categorical variables
• Feature scaling and normalization
These operations improve dataset quality and reduce analytical errors.

8.4 Encoding of Categorical Variables

Machine learning algorithms cannot directly process textual categorical values.
Therefore, encoding techniques are used to convert reservation categories into numerical values.
Examples include:

- Label Encoding
- One-Hot Encoding

Encoding improves compatibility between hotel reservation datasets and machine learning algorithms.

8.5 Feature Scaling

Feature scaling normalizes numerical reservation data into a consistent range.
This prevents features with larger numerical values from dominating smaller features.

Advantages of scaling:

- Improved model convergence
- Better prediction accuracy
- Faster computation
- Balanced feature contribution

8.6 Train-Test Split

The dataset was divided into training and testing sets.
This is an important stage in machine learning workflows because:

- Training data is used to train the prediction model.
- Testing data is used to evaluate model performance.

The split helps prevent overfitting and improves model generalization for hotel reservation cancellation prediction.

8.7 Role of EDA in Machine Learning

Exploratory Data Analysis plays a major role in machine learning preparation.
EDA helps identify:

- Correlation between reservation-related features
- Outliers and anomalies
- Data imbalance
- Statistical patterns
- Customer booking trends
- Reservation cancellation behavior

8.8 Machine Learning Workflow

The complete machine learning workflow followed in this project includes:

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis
4. Feature Engineering
5. Feature Scaling
6. Data Splitting
7. Machine Learning Model Training
8. Model Evaluation
9. Visualization and Interpretation
10. Reservation Cancellation Prediction

8.9 Observations and Results

The preprocessing and feature engineering process significantly improved dataset quality.
Key observations:

• The cleaned dataset became more structured and consistent.
• Visualization improved understanding of customer booking trends.
• Feature engineering enhanced machine learning readiness.
• Proper preprocessing reduced analytical complexity.
• Reservation patterns and cancellation behavior became easier to analyze.

The project demonstrates how machine learning preparation techniques improve data science workflows and analytical performance in hotel reservation analytics.

8.9 Future Scope in Machine Learning

Future improvements may include:

• Predictive hotel reservation analytics
• Reservation cancellation prediction
• Dynamic pricing prediction systems
• Deep learning models
• Dashboard deployment using Power BI or Tableau
• Real-time hotel reservation analytics systems

Machine learning can significantly improve organizational decision-making and workforce planning.

9. Reference

https://drive.google.com/file/d/19vY2goE0SZRjJQZFxxyjR-nQvQcCDcKk/view?usp=sharing

Linkdin-https://www.linkedin.com/posts/abhinavkumar9211_machinelearning-datascience-python-ugcPost7462928272666902528-Rddi?utm_source=share&utm_medium=member_desktop&rcm=ACoAACnsPY4BxQcUTvFNUydHCgkJ1oJmdBbRSo

Github-https://github.com/abhinakumar/Hotel-Booking-Cancellation-Analysis